

ASSIGNMENT 5

QUESTION 1

Build a Convolutional Neural Network to predict digits using the MNIST datasets

SOLUTION

```
import tensorflow as tf
from tensorflow.keras import datasets, layers, models

(x_train, y_train), (x_test, y_test) = datasets.mnist.load_data()

x_train = x_train.reshape((x_train.shape[0], 28, 28, 1))
x_test = x_test.reshape((x_test.shape[0], 28, 28, 1))

x_train, x_test = x_train / 255.0, x_test / 255.0

model = models.Sequential([
    layers.Conv2D(32, (3,3), activation='relu', input_shape=(28, 28, 1)),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3,3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])

model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

model.fit(
    x_train, y_train, epochs=5, batch_size=64, validation_split=0.1
)

test_loss, test_acc = model.evaluate(x_test, y_test, verbose=2)
print('\nTest accuracy:', test_acc)
```

OUTPUT

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ————— 3s 0us/step
Epoch 1/5
2025-09-02 08:59:27.902675: W external/local_xla/xla/tsl/framework/cpu_allocator_impl.cc:84]
Allocation of 169344000 exceeds 10% of free system memory.
844/844 ————— 22s 24ms/step - accuracy: 0.9458 - loss: 0.1785 - val_accuracy:
0.9833 - val_loss: 0.0576
Epoch 2/5
844/844 ————— 19s 23ms/step - accuracy: 0.9821 - loss: 0.0559 - val_accuracy:
0.9878 - val_loss: 0.0454
Epoch 3/5
844/844 ————— 21s 23ms/step - accuracy: 0.9881 - loss: 0.0380 - val_accuracy:
0.9853 - val_loss: 0.0528
Epoch 4/5
844/844 ————— 19s 23ms/step - accuracy: 0.9906 - loss: 0.0295 - val_accuracy:
```

0.9895 - val_loss: 0.0392

Epoch 5/5

844/844  19s 23ms/step - accuracy: 0.9925 - loss: 0.0235 - val_accuracy:

0.9897 - val_loss: 0.0365

313/313 - 1s - 4ms/step - accuracy: 0.9889 - loss: 0.0361

Test accuracy: 0.9889000058174133